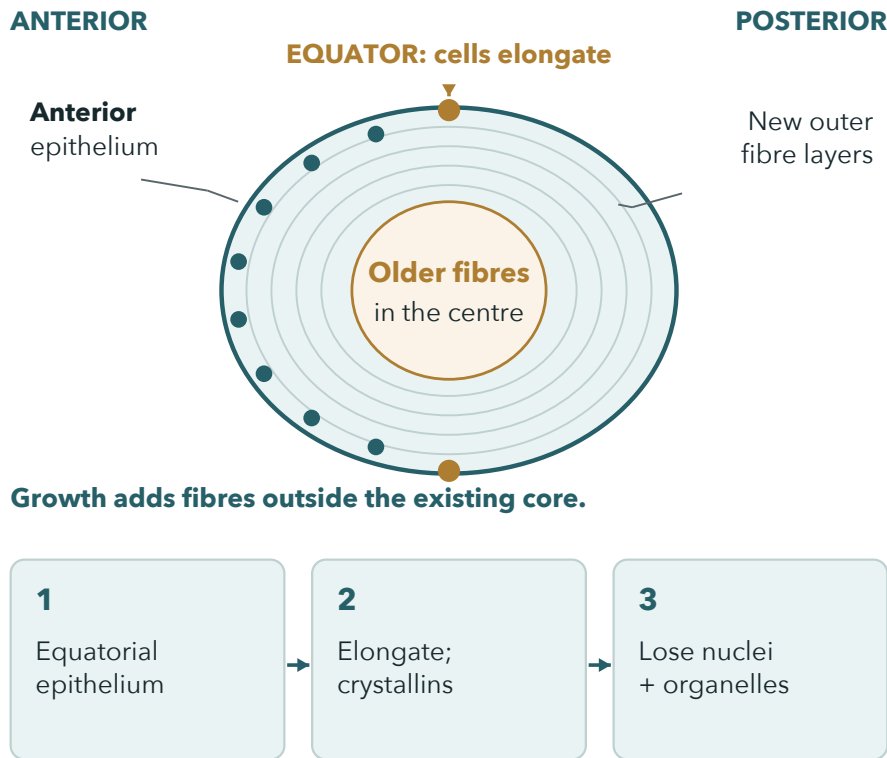


Lens fibres and capsule

Two structures, two jobs: clear fibre packing inside a basement-membrane envelope.

New fibres, older centre

Where fibres come from



Section through the lens; schematic proportions. New secondary fibres are added outside older fibres.

Read the specimen

Finding	Inference
Anterior monolayer	Living lens epithelium; proliferative cells contribute at the equator.
Equatorial elongation	Secondary fibre differentiation and crystallin accumulation.
Mature fibre centre	Nuclei and organelles are lost; older fibres remain centrally.

One exam distinction

Primary fibres arise from posterior lens-vesicle cells during embryogenesis. Secondary fibres arise from equatorial epithelial differentiation throughout life.

The capsule is not a cell layer

The adult lens envelope

ANTERIOR / AQUEOUS

POSTERIOR / VITREOUS

Epithelium

anterior only

No posterior
epithelial sheet

Epithelium

anterior only

Lens fibres

avascular

Nutrients

Wastes

CAPSULE = BASEMENT MEMBRANE

Continuous envelope around anterior and posterior poles.

The capsule surrounds both poles. An epithelial sheet lies beneath its anterior part, not its posterior part.

Separate the structures

Structure	High-yield role
Capsule	Specialised basement membrane; supports and encloses the lens.
Epithelium	Metabolism, ion transport and new fibre production.
Mature fibres	Crystallin-rich optical material with few scattering organelles.

Nutrition without vessels

The adult lens is avascular. Nutrients, especially from aqueous humour, and wastes exchange with surrounding ocular fluid across the capsule and surface cells.

Sources NIH Webvision: Lens | NIH: Eye embryology